Electron Configuration

Answer key with worked solutions

A. Understanding

1. p subshell.

> **3 orbitals**, each holding 2 electrons: **6 electrons** in total.

2. Pauli and Hund.

> **Pauli:** an orbital holds at most two electrons, and they must have opposite spins. **Hund:** electrons in the same subshell occupy separate orbitals with parallel spins before any orbital is doubly occupied.

B. Atoms

3. Full configurations.

> (a) C: $1s^2\ 2s^2\ 2p^2$ (b) Mg: $1s^2\ 2s^2\ 2p^6\ 3s^2$ (c) Cl: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

4. Noble-gas configurations.

> (a) Ca: **[Ar] $4s^2$** (b) Fe: **[Ar] $3d^6\ 4s^2$** (c) Br: **[Ar] $3d^{10}\ 4s^2\ 4p^5$**

5. Identify the elements.

> (a) 7 electrons: **nitrogen** (b) 10 + 7 = 17: **chlorine** (c) 18 + 14 = 32: **germanium**

C. Orbital diagrams

6. Nitrogen.

> 2s: [↑↓] 2p: [↑] [↑] [↑]. By Hund's rule the three 2p electrons occupy separate orbitals: **3 unpaired electrons**.

7. Unpaired electrons.

> (a) S, [Ne] $3s^2\ 3p^4$: 3p is [↑↓] [↑] [↑], so **2** (b) Ne, $1s^2\ 2s^2\ 2p^6$: all subshells full, so **0**

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D. Ions

8. Ions.

> (a) O^{2-}: 8 + 2 = 10 electrons, $1s^2\,2s^2\,2p^6$ = [Ne] (b) Mg^{2+}: 12 − 2 = 10 electrons, [Ne] (c) Fe^{3+}: remove both 4s electrons, then one 3d: **[Ar] 3d⁵**. O^{2-} **and** Mg^{2+} have the same configuration (they are isoelectronic with neon).

E. Challenge

9. Chromium.

> A half-filled d subshell (one electron in each of the five 3d orbitals) is especially stable, so one 4s electron moves into 3d. Unpaired electrons: five in 3d plus one in 4s = **6**.

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